Lenses and image formation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use ray directions to explain converging and diverging lens behavior.

SUCCESS CRITERIA - I can...

1. Distinguish converging and diverging lenses.
2. Trace principal rays with a clear optical axis.
3. State whether an image is real or virtual in the given case.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “lens” and “focal point” to explain your model or result. Sentence starters are optional.

lens: a transparent element that changes light-ray directions

focal point: where suitable parallel rays meet or appear to originate

ray: a line showing the modeled direction light travels

01 Bring back an earlier idea

A source approaching a stationary listener at 15 m/s emits 500 Hz.
Use $v_{sound}=340$ and $f_{obs}=f_s\ v/(v-v_s)$.
Find observed frequency.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $1/f = 1/d_o + 1/d_i$.
Thin lens: f is focal length; o = object, i = image. Use the stated sign convention and a ray diagram.

Worked reasoning: A converging lens can form a real image of an object beyond its focal point. A ray parallel to the axis passes through the far focus; a central ray is approximately undeviated in the thin-lens model.

Your turn: Draw these two rays and mark their intersection for an object beyond 2 f.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Complete ray sketches for parallel rays through a converging lens and a diverging lens.
Label the optical axis.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Left: converging lens. Right: diverging lens. Horizontal lines mark the optical axes.

Marzano check - circle: ① ② ③

04 Solve and justify

A converging lens has f=10 cm and object distance 30 cm.
Find image distance and magnification $m=-d_i/d_o$.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A magnifying glass produces an upright virtual image when object is inside f.
Explain why “converging lens always makes a real image”is wrong.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

06 Choose a method

A wave snapshot has crests at 0.2, 0.6, 1 m.
Find wavelength. Its period is 0.06 s; find frequency.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

07 Exit check

For a converging lens f=10 cm and d_o=5 cm, find d_i and m.
State image kind.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

A lens has f=10 cm and object distance 30 cm.
Use $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$.
Find image distance and magnification.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③